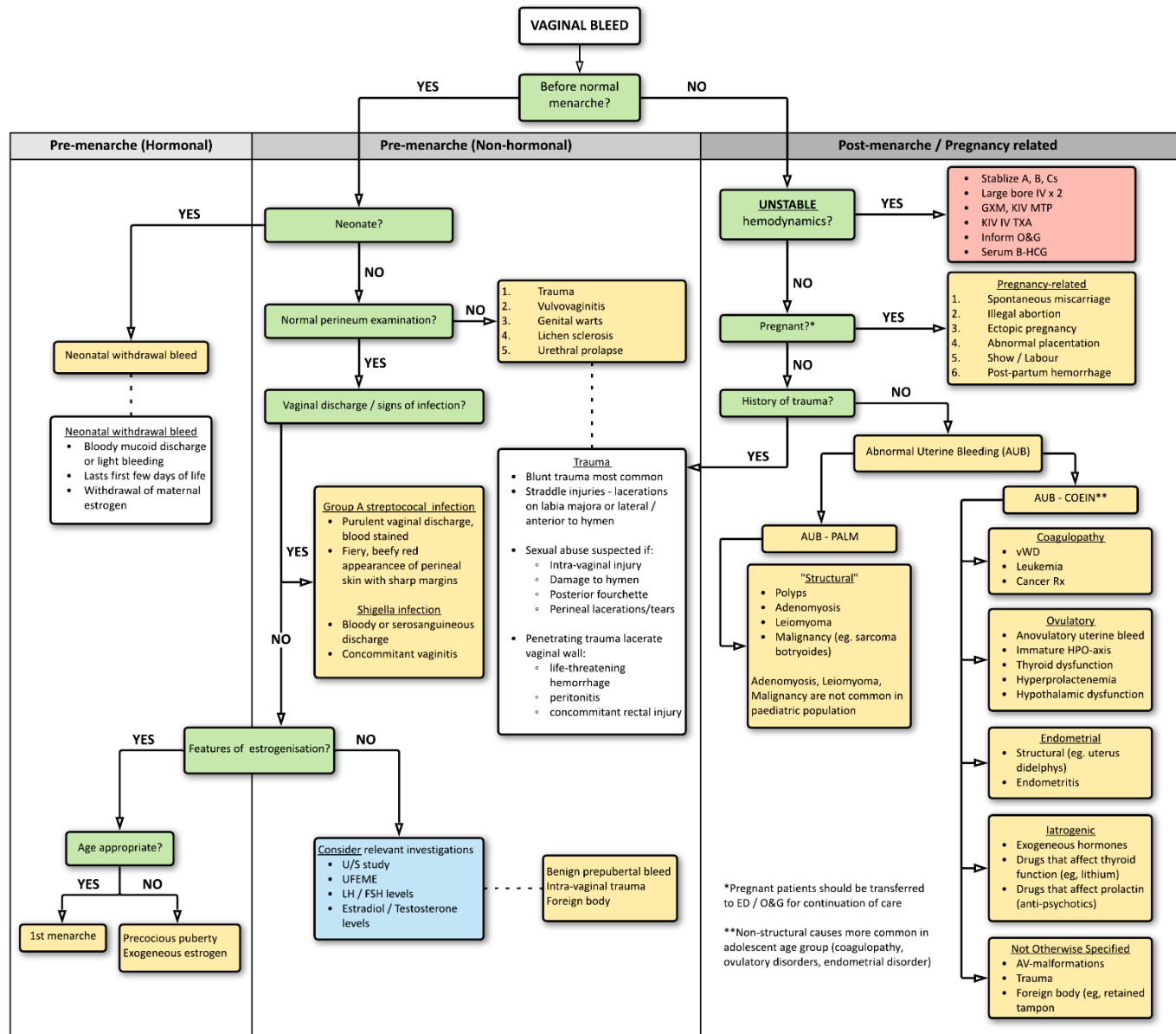


GYNAECOLOGICAL PROBLEMS IN THE CHILDREN'S EMERGENCY

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1 APPROACH TO VAGINAL BLEEDING IN THE CHILDREN'S EMERGENCY



2 ABNORMAL UTERINE BLEEDING (AUB)

2.1 CLASSIFICATION OF ABNORMAL UTERINE BLEEDING

- 4 parameters: Frequency, Regularity, Duration & Volume

PARAMETER	DESCRIPTION	VALUES	
		ADULTS	ADOLESCENTS
Frequency (cycle duration)	Infrequent Normal Frequent	> 35 days 21 – 35 days < 21 days	> 45 days 21 – 45 days < 21 days
Regularity (inter-cycle variation)	Regular Irregular	Variation ≤ 7 to 9 days Variation > 7 to 9 days	Not defined
Duration	Normal Prolonged	2 to 7 days > 8 days	≤ 7 days > 7 days
Volume (subjective)	Heavy	- Fully soaked pad after 2h or 4 soaked pads per day for ≥ 3 days - Passing clots (> 2.5cm) - Accidents (in the day/night) - Heavy menstrual bleeding that affects quality of life (physical, emotional, social, and/or material)	

A) 1 cycle = D1 of menses to D1 of next menses

B) Over the past 6 months

2.2 ETIOLOGY OF AUB [PALM-COEIN CLASSIFICATION]

- Adolescents are more likely to have non-structural etiologies than structural

Structural

P	Polyp		U/S Hysteroscopy Histopathology
A	Adenomyosis	Endometrial invasion of the myometrium	U/S study Histopathology
L	Leiomyoma	Benign tumors of the smooth muscle	U/S study to sub-classify or MRI
M	Malignancy & hyperplasia	Endometrial hyperplasia with cytological atypia & carcinoma (rare in CE)	U/S study Histology

Non-Structural

C	Coagulopathy	vWD (most common) Leukaemia Cancer Rx	PT/PTT/INR vWF:Ag, vWF:RCO
O	Ovulatory dysfunction (irregular menses)	Immature HPO-axis Anovulatory uterine bleeding Thyroid dysfunction Hyperprolactinemia Hypothalamic dysfunction	Menstrual calendar Amenorrhoea panel or Female Hormone panel Adrenal hormones
E	Endometrial dysfunction (volume of flow)	Increased surface area (structural) – uterus didelphys Endometritis	U/S study Swabs Histopathology
I	Iatrogenic	Exogenous hormones, drugs that affect TFT, PRL, TCM	Medical history Drug history Hormone panel
N	Not otherwise specified	Eg. AV-malformations, trauma, foreign body	U/S or other imaging Hysteroscopy

3 OVERVIEW OF ABNORMAL UTERINE BLEED IN ADOLESCENTS

3.1 ANOVULATORY VS. OVULATORY CYCLES

ANOVLATORY CYCLES	OVULATORY CYCLES
▫ No corpus luteal formation → progesterone is not produced ▫ Endometrium continues to proliferate under influence of unopposed estrogen ▫ Endometrium is shed → prolonged and heavy	▫ Corpus luteum produces progesterone ▫ Progesterone converts estrogen primed proliferative endometrium to secretory endometrium

3.2 CAUSES OF ABNORMAL UTERINE BLEED

3.2.1 Irregular Menstrual Bleed

3.2.1.1 Immature HPO-axis

- In adolescent girls, there is failure to mount an LH surge → corpus luteum is not formed → low progesterone
- A history of irregular and heavy uterine bleed is common as part of pubertal transition
- Treatment is with progesterone if endometrium is thick (unopposed estrogen action), estrogen and progesterone if endometrium is thin (hypoestrogenic)

3.2.2 Heavy Menstrual Bleed

3.2.2.1 Bleeding disorders

- Platelet disorders (primary vs. acquired) – most commonly von Willebrand Disease (vWD)
- Coagulopathy – clotting factor deficiency
- Cancer and its treatment

3.2.2.2 Others:

- Systemic diseases that affect ovarian or liver function resulting in abnormal ovulation (eg. DM) or coagulation (eg. SLE, renal failure, hematologic dysplasia)
- Endocrine disorders (hyperthyroidism or hypothyroidism)
- Structural (eg. polyps, fibroids)
- Medications (eg. anticoagulants, platelet inhibitors, TCM)

3.2.3 Polycystic Ovary Syndrome (PCOS)

- Features of PCOS overlap with expected changes in adolescent patients during pubertal transition
- Patients with features of PCOS could be labelled “at increased risk” with reassessment at full reproductive maturity i.e. 8-years post-menarche

FEATURES OF PCOS	
Ovulatory dysfunction presenting as irregular menstrual cycles	Irregular cycles and ovulatory dysfunction secondary to HPO axis immaturity is normal in an adolescent
Hyperandrogenism	No norms for testosterone levels in adolescents Acne vulgaris is not uncommon in adolescents, hirsutism differs by ethnicity
Polycystic ovaries	Trans-abdominal scan is usually performed → difficulty in differentiating polycystic vs. multi-cystic ovaries which can be normal

3.2.4 Amenorrhea

- **Primary:** Lack of menarche by 15-years of age **OR** by 3-years after thelarche
Usually the result of a genetic or anatomical abnormality
- **Secondary:** Absence of menses for > 90 days, after previously menstruating

CAUSES OF AMENORRHEA	
BY SYSTEM	MOST COMMON CAUSES IN ADOLESCENTS
EXCLUDE PREGNANCY	
<u>Hypothalamic & pituitary disease</u> a. Constitutional pubertal delay b. Functional hypothalamic amenorrhea c. Isolated GnRH deficiency d. Hyperprolactinemia e. Infiltrative disease f. Systemic disease <u>Gonadal dysgenesis / Primary Ovarian Insufficiency</u> a. Turner syndrome b. 46 XY gonadal dysgenesis <u>PCOS</u> <u>Outflow Tract Disorders</u> a. Imperforate hymen b. Transverse vaginal septum c. Mullerian agenesis <u>Receptor abnormalities & enzyme deficiencies</u> a. Complete androgen insensitivity syndrome b. 5-alpha-reductase deficiency c. 17-hydroxylase deficiency d. Estrogen resistance	a. Gonadal dysgenesis (eg. Turner's syndrome) b. Mullerian agenesis (absence of vagina, sometimes with absence of uterus) c. Physiologic pubertal delay – constitutional delay of puberty, chronic systemic disease, acute illness) d. Eating disorder e. Hypopituitarism

4 MANAGEMENT OF ABNORMAL UTERINE BLEEDING

4.1 ACUTE MANAGEMENT

- Stabilize patient's ABCs
- Clinical assessment of anaemia
 - Symptoms of anaemia
 - Physical signs of anaemia
- Low threshold to perform FBC (clinical assessment might not correlate with actual Hb level)
- If indicated, consider: Reticulocyte count, thyroid function test, GXM, PT/PTT/INR, Iron panel)

PHYSICAL SIGN	ESTIMATED HB
Conjunctival pallor	10 g/dL
Hemic murmur	8 g/dL
Palmar pallor	7 g/dL
Eye ball bruit	5 g/dL

4.2 INDICATIONS FOR O&G REFERRAL

- An O&G consult should be obtained for patients with the following:
 - Unstable patients
 - Patients with anaemia (proven on FBC)
 - Heavy on-going bleed
 - Bleeding for ≥ 10 days
- Should clinical need arise, consult CE senior regarding further discussion:
 - Dr Nau'shil Kaur Randhawa, Associate Consultant
 - Dr Susan Logan, Senior Consultant

4.3 ACUTE CONTROL OF BLEEDING

- Haemostatic Therapy
 - PO tranexamic acid **15 – 25mg/kg** (1g – 1.5g) 8H for ≤ 5 days (Started on discharge for on-going bleeds OR D1 of next cycle)
 - IV tranexamic acid **10mg/kg** (1g) once (if unable to tolerate orally)
- If already on tranexamic acid, consider O&G referral for hormonal therapy

4.4 IRON SUPPLEMENTATION

- Iron deficiency is often the result of prolonged AUB compounded by dietary risk factors (e.g., vegetarian)
- Dose of supplementation = **4 – 6 mg/kg (BW)** per day of elemental Fe
- Expected rise in Hb = 0.5 – 1 g/dL per week

IRON FORMULATION	COMPONENTS / ELEMENTAL IRON
Maltofer 100mg tablet	100% elemental iron = 100mg
Ferrous Gluconate Co Cap (Sangobion)	12% elemental iron = 30mg (Vitamin B12, folic acid, ascorbic acid, manganese, copper sulphate, sorbitol)
Iberet Folate SR Tablet	20% elemental iron = 105mg (Thiamine, Vitamin B12, B3, B5, B6, B12, folic acid, calcium panthothenate, ascorbic acid)

4.5 DISPOSITION

- Indications for admission:
 - Hemodynamic instability (tachycardia, hypotension, orthostatic vital signs)
 - Severe anaemia:
 - Hb < 7 g/dL or
 - Hb < 10 g/dL with on-going bleed and/or symptomatic
 - On-going heavy bleeding despite tranexamic acid and/or hormonal therapy
- If for discharge, arrange TCU with O&G Adolescent clinic 1 – 2 weeks after discharge.

5 OVERVIEW OF THERAPEUTIC OPTIONS

5.1 HORMONE THERAPY

5.1.1 Combination oral contraceptive (COC) pills

- Ethinyl estradiol (EE) 35mcg or less
- Dosing regimens of COC:
 1. Monophasic vs. Multiphasic
 - i. Monophasic – active pills contain same dose and ratio of EE and progestin
 - ii. Multiphasic – active pills contain varying doses and ratios within a cycle (not available in NUH)
 2. Cyclic dosing
 - i. Typical: 21/7 (active/placebo) in a 28-day cycle
 - ii. Alternative: 24/4 → shorter hormone free days with reduced risk of breakthrough folliculogenesis and side effects of hormone withdrawal (mood symptoms, pelvic pains, and headaches)
 3. Extended/continuous dosing
 - i. Extended – prolonged suppression of menstruation with withdrawal bleeds occurring 1 week per 2 or 3 months based on cycle regime
 - ii. Flexible – COC pill taken every day, if break through bleeding for 3 – 4 days continuous then stop for 3 – 4 days
- Risks of COC:
 1. VTE risk
 2. Cardiovascular risk including stroke
 3. Bones – decreased BMD in 20 mcg (EE) preparations
- These risks do not normally apply to adolescents unless specific medical conditions. Please refer to WHO [“Medical eligibility criteria for contraceptive use \(Fifth Edition 2015\)”](#)
- Examples of COC:

COC*	ESTROGEN (EE)	PROGESTIN
Microgynon 30	30 mcg	Levonorgestrel
Diane-35	35 mcg	Cyproterone
Yaz/Yasmin	20/30 mcg	Drospirenone
Mercilon	20 mcg	Desogestrel

**Names may be different.*

5.1.2 Progestin-only methods

- For patients with contraindications to estrogen or avoiding estrogen-exposure
- Formulations:
 1. Long-acting contraceptives e.g., Levonorgestrel Intra-Uterine System (IUS) or Etonogestrel implant
 2. Depot medroxyprogesterone acetate (DMPA) injection
 3. Progestin-Only Pills: norethisterone

5.2 NON-HORMONAL THERAPY

- Tranexamic acid
 - Reduces blood loss by 25 – 50%
 - Taken for up to 5 days during menses
 - Caution in kidney disease
- NSAIDs
 - Blocks prostacyclin production in endometrium → vasoconstriction and reduced bleeding
 - Helps with dysmenorrhea

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